

PriceShield

Data analytics for Food Security in Gambia



Food Security in Gambia

Price Volatility



Stable Prices



Problems



Shortage



Drought



Oil Price



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(The Gambia)

Actions



Regulate

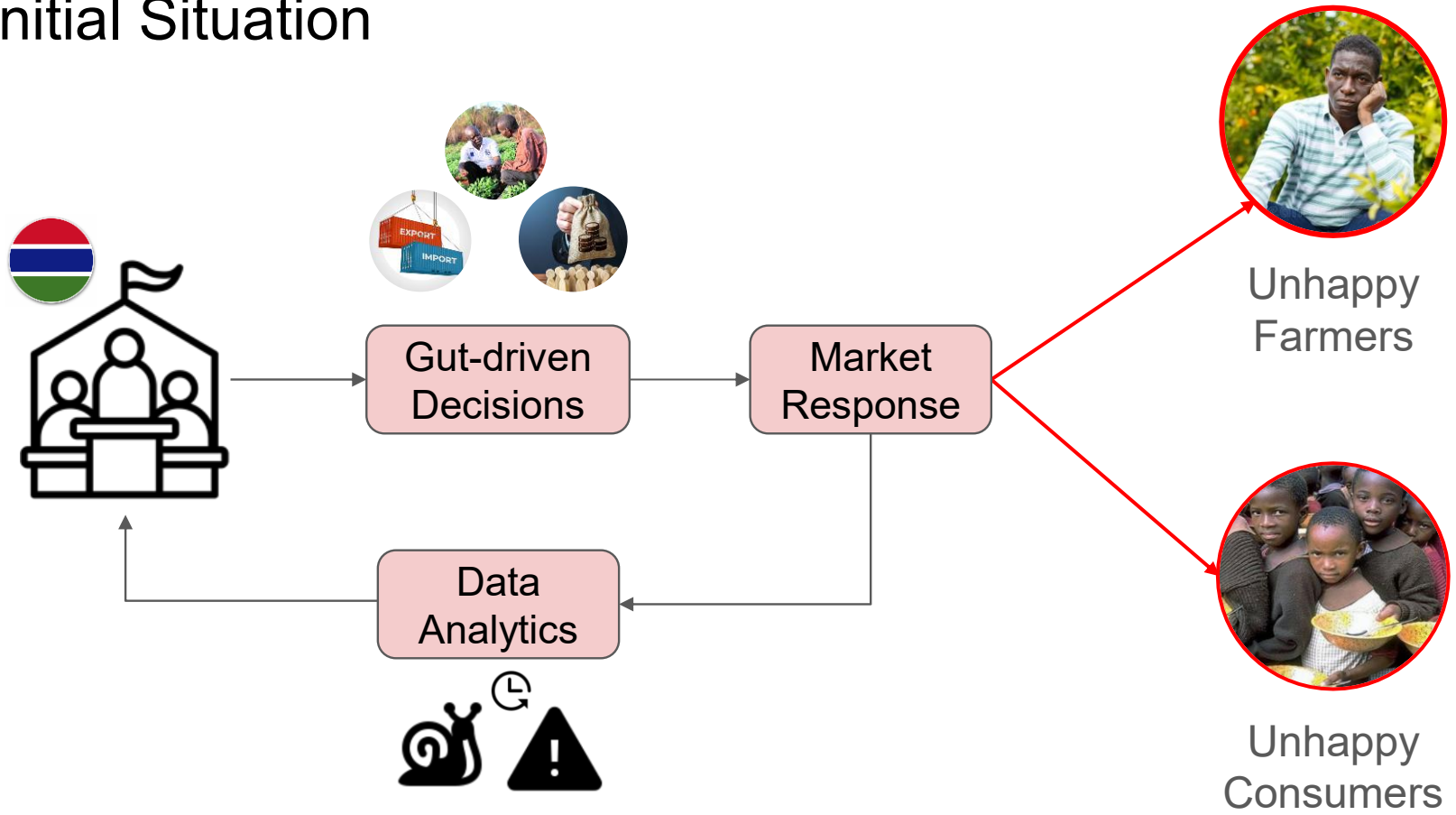


Train

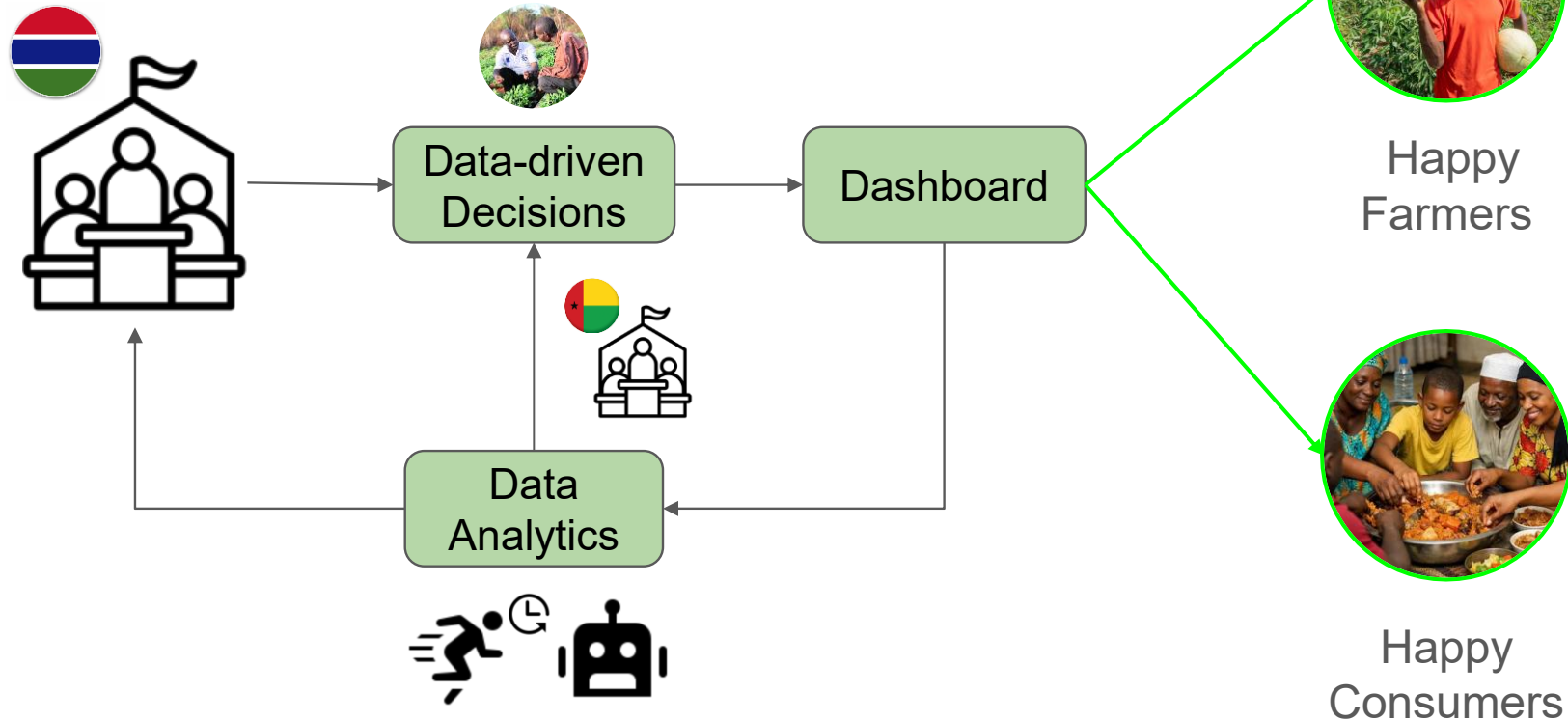


Subsidy

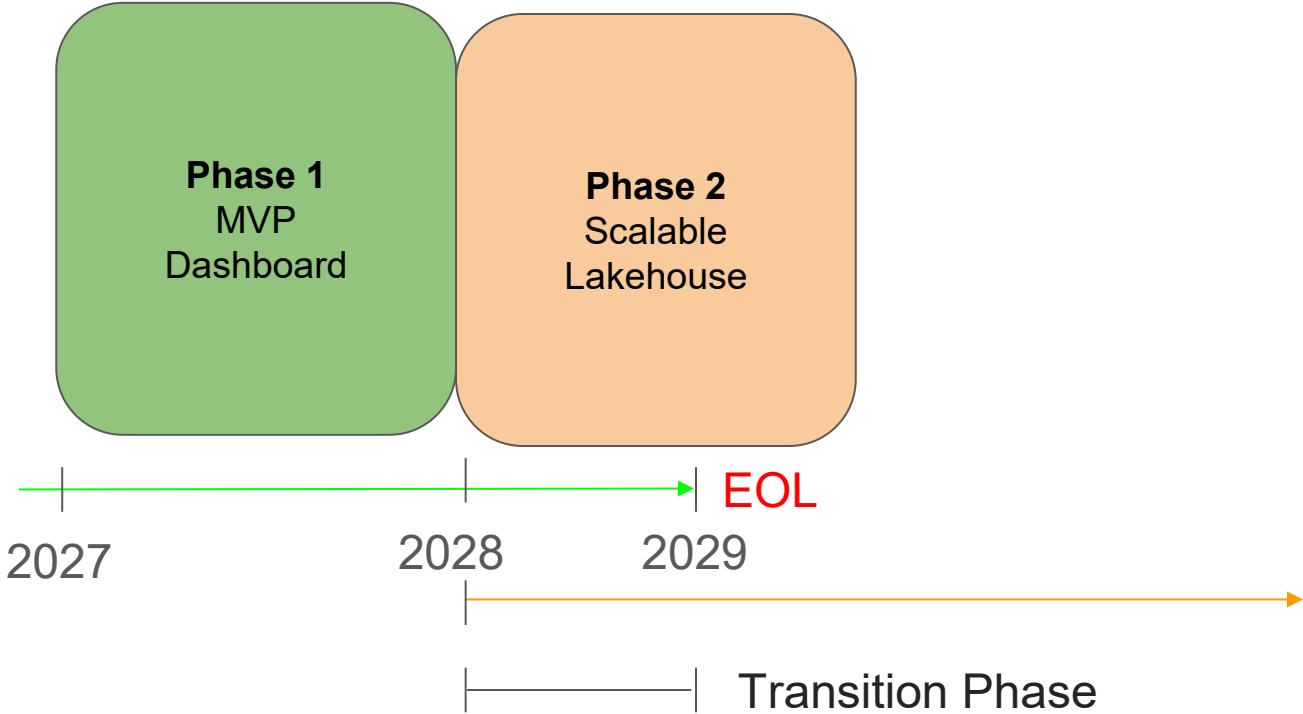
Initial Situation



Solution



Dual-Roll Strategy



System Features & Functional Requirements

System feature	Business purpose
1 ADP - Automated Data Processing	Automatically imports, stores and processes new monthly food-price data without manual Excel work.
2 RCD - Reliable and Comparable Data	Validates and standardises food-price data so that commodities, units and currencies can be compared reliably.
3 MON - Price Monitoring and Country Comparison	Shows national and regional price trends and compares matching commodities between The Gambia and Guinea-Bissau.
4 FEW - Forecasting and Early Warning	Forecasts future commodity prices and warns about significant expected price increases.
5 REP - Monitoring and Reporting	Provides authorised analysts with a secure Power BI report for fast meeting analysis and controlled reporting to government decision-makers.

System Features & Functional Requirements

REQ-ADP-04 Processing deadline

The system shall complete the monthly processing before the updated dashboard is published. Compliance is achieved when at least 95% of valid monthly deliveries are published within 24 hours of arrival.

REQ-RCD-05 Currency standardisation

The system shall convert comparable prices from Gambian dalasi and West African CFA francs into United States dollars using an approved monthly exchange rate. Compliance is achieved when each converted value retains its original price, original currency, applied exchange rate, exchange-rate date and converted USD value.

REQ-REP-04 Report performance

The Power BI report shall load pages and update visuals within five seconds for at least 95% of interactions. After extended inactivity, the first request may take up to 30 seconds due to server reactivation.

System Features & Functional Requirements

PriceShield

System Features and Requirements

Data Intelligence for Food Security in The Gambia

Document structure

Five business-oriented system features with concise descriptions, use case flows and measurable functional requirements.

[Link to Document](#)

1 Automated Data Processing

Description

Automatically imports, stores and processes new monthly food-price data without manual Excel work.

Use Case Flow

1. A new monthly food-price file becomes available.
2. PriceShield retrieves and stores the original file.
3. The system keeps the records for The Gambia and Guinea-Bissau.
4. The system starts validation and transformation.
5. If processing succeeds, the system publishes the new data.
6. If processing fails, the system keeps the previous valid dataset and informs Fatou.

Functional Requirements

REQ-ADP-01 Automated ingestion

The system shall automatically ingest every new monthly food-price dataset. Compliance is achieved when all valid monthly deliveries are processed without manual file upload.

REQ-ADP-02 Raw-data preservation

The system shall store every received source file unchanged before processing. Compliance is achieved when each file has a unique file ID, ingestion timestamp and source identifier.

REQ-ADP-03 Country filtering

The system shall retain only observations for The Gambia and Guinea-Bissau in the curated analytical dataset. Compliance is achieved when the curated dataset contains no records from other countries.

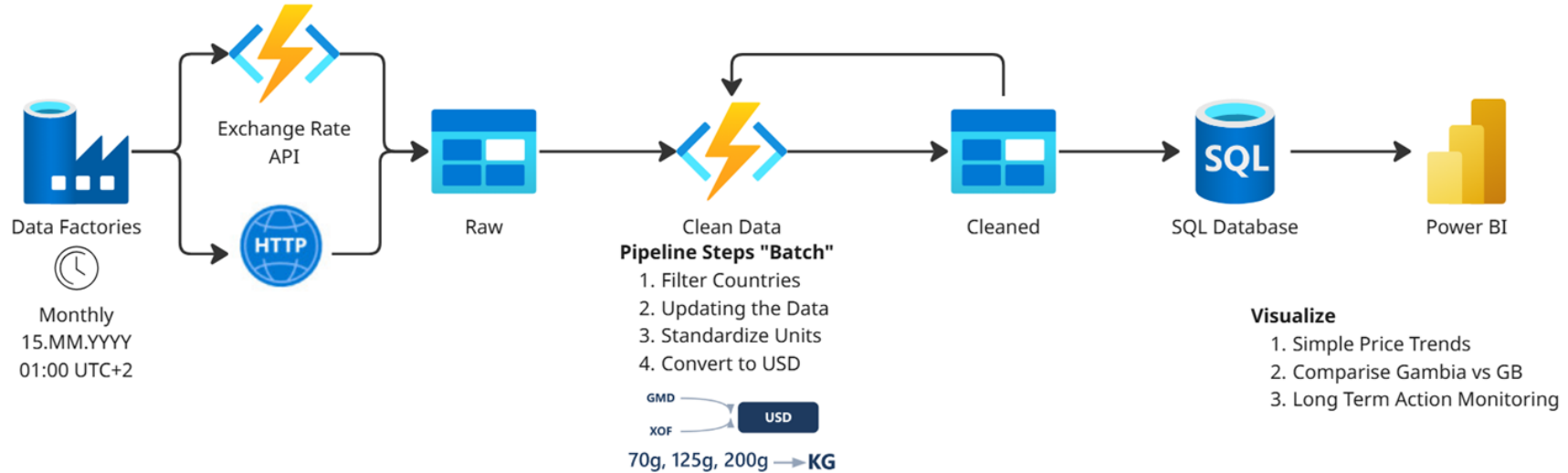
REQ-ADP-04 Processing deadline

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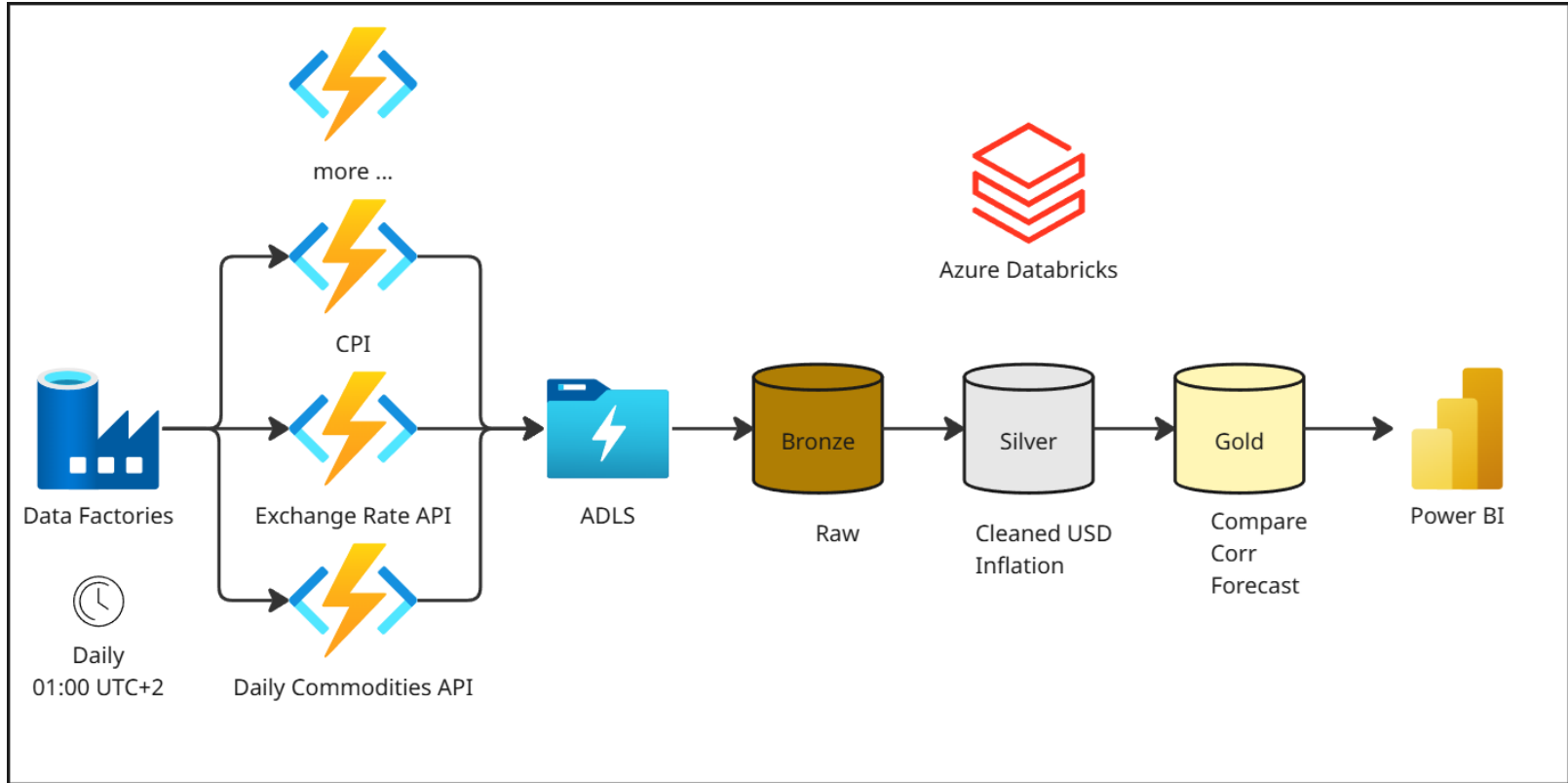
REQ-ADP-05 Failed-load protection

The system shall preserve the latest validated dataset when a processing run fails. Compliance is achieved when no failed delivery replaces the current production data.

Phase 1 : MVP (Serverless Architecture)



Phase 2 : Scalable Lakehouse (Databricks)



Seasonal Price Analysis

Price Change (vs Last Month)

%0,51

2021-07 vs 2021-06

Price Change (vs Last Year)

%13,13

2021-07 vs 2020-07

Gambia vs Guinea-Bissau Price Gap

-%6,76

Total Price Records

45K

Country

☒ Gambia

☒ Guinea-Bissau

Gambia Regions

All

Guinea-Bissau Regions

All

Year

2016

2021

Market

All

Commodity

All

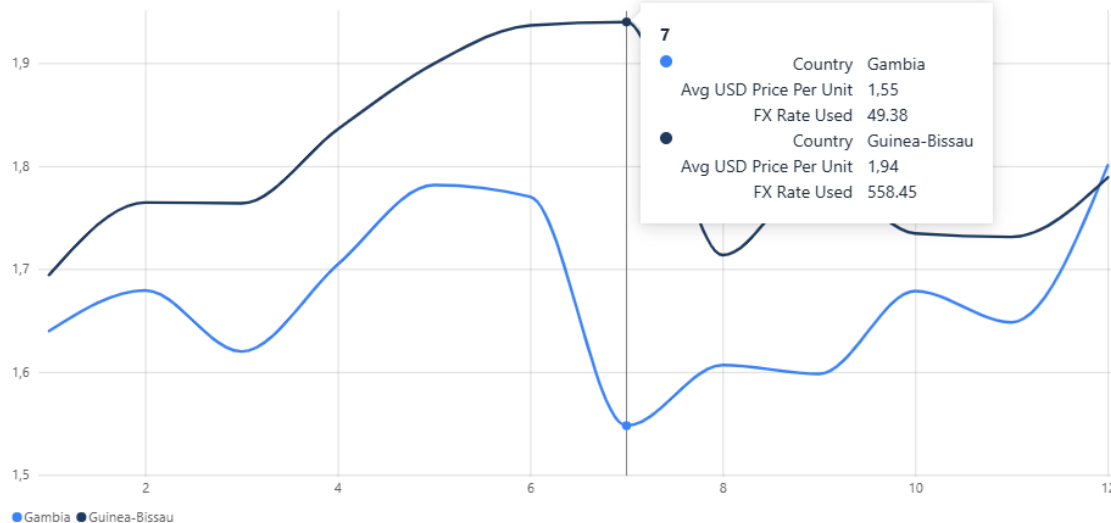
Unit Category

KG

Price Type

Retail

Avg USD Price Per Unit and FX Rate Used
by Month and Country



Most Recent Data Month

01.07.2021

Pipeline Last Updated

20.09.2026 17:01

% Missing Months (Gambia)

%4,55

% Missing Months (Guinea-Bissau)

%0,00

Price Trend Over Time

Price Change (vs Last Month)

%0,51

2021-07 vs 2021-06

Price Change (vs Last Year)

%13,13

2021-07 vs 2020-07

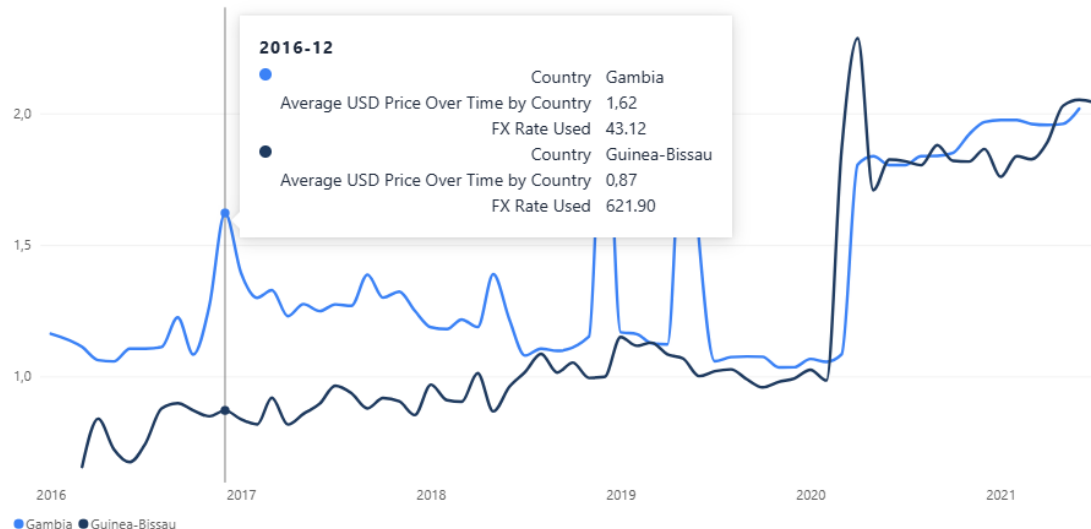
Gambia vs Guinea-Bissau Price Gap

-%6,76

Total Price Records

45K

Average USD Price Over Time by Country and FX Rate Used
by ObservationDate and Country



Country

☒ Gambia

☒ Guinea-Bissau

Gambia Regions

All

Guinea-Bissau Regions

All

Year

2016

2021

Market

All

Commodity

All

Unit Category

KG

Price Type

Retail

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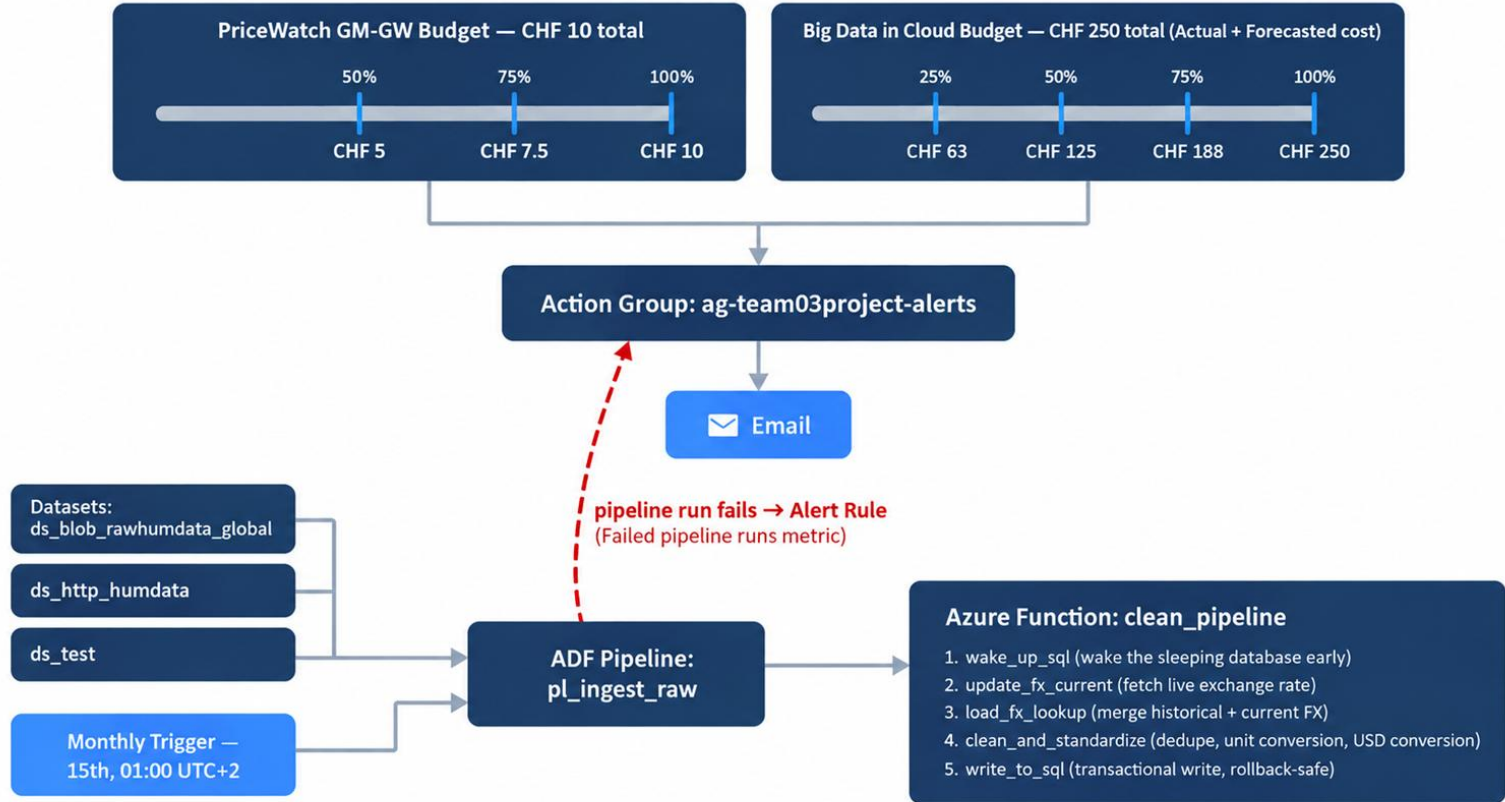
% Missing Months (Gambia)

%4,55

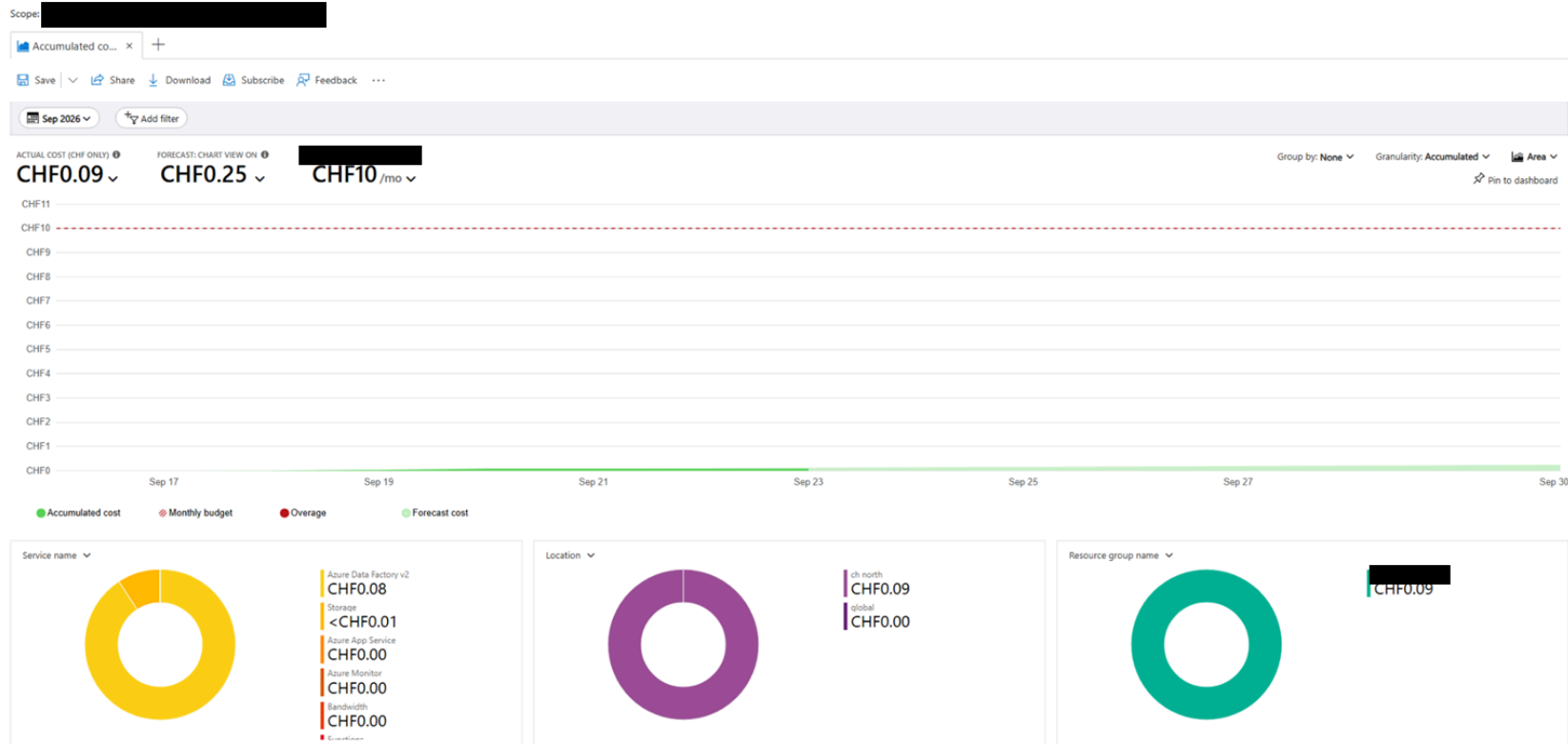
% Missing Months (Guinea-Bissau)

%0,00

Azure Portal Summary



What We Spend in Development



Phase 1: Prototype Cost Breakdown & Sizing

Total Monthly Cost: \$32.57 / month (\$4.57 Azure Infrastructure + \$28.00 Power BI Pro)

Service Component	Monthly Cost	Capacity Sizing Parameters	
Azure Data Factory 	\$1.26	1x1000 Activity Run, 1 Data Movement Unit (DMU)	Sized for batch-scheduled runs to ingest external files (110 MB CSV).
Azure Functions 	\$0.00	Flex Consumption, 2048 MB RAM, 1x10 Executions/mo	Handles API requests (exchange rates); stays inside the Azure free tier.
Azure Blob Storage 	\$0.55	General Purpose v2, Hot Tier, 20 GB Capacity	Provisioned 20 GB raw storage, ample headroom ~15 years of data.
Azure SQL Database 	\$2.76	Serverless vCore (Gen 5), 1 billed vCore	Auto-pauses when inactive, so compute is only charged during active queries.
Power BI Pro 	\$28.00	2 Subscriptions	1 authoring license required for dashboard creation and 1 for consumption.

Phase 2: Scalable Lakehouse Cost Breakdown & Upgrades

Total Monthly Cost: \$39.26 / month (\$11.26 Azure Infrastructure + \$28.00 Power BI Pro)

Service Component	Monthly Cost	Capacity Sizing Parameters	
Azure Data Factory 	\$1.26	1x1000 Activity Run, 1 Data Movement Unit (DMU)	Kept identical at 1 batch pipeline run per month.
Azure Functions 	\$0.00	Flex Consumption, 2048 MB RAM, 1x10 Executions/mo	Kept identical; remains inside Azure free grant.
ADLS Gen2 Storage 	\$2.14	Standard ADLS Gen2, Hot Tier, 20 GB + High-Priority Ops	Switched to Data Lake Gen2. Includes high-volume analytics operation allowances.
Azure Databricks 	\$7.86	Jobs Compute, Premium, 1 D3v2 Node (4 vCPUs, 14 GB RAM)	Sized for 15 hours of active Spark transformations per month. Spark parallel computing for data processing.
Power BI Pro 	\$28.00	2 Subscriptions	Kept identical; 1 authoring license required for dashboard creation and 1 for consumption. +\$14 for every new seat

Side-by-Side Cost & Capacity Comparison

Transitioning from prototype to scalable lakehouse, adding +\$6.69/month in net infrastructure

Category / Service	Phase 1: Prototype	Phase 2: Scalable	Capacity Sizing Delta
Ingestion (ADF + Functions)	\$1.26	\$1.26	Identical: 1 activity run, 1 DMU, API executions.
Storage Engine	\$0.55 (Blob Storage)	\$2.14 (ADLS Gen2)	+\$1.59 for Data Lake analytics operations allowance.
Compute / Processing	\$2.76 (SQL Serverless)	\$7.86 (Databricks Jobs)	+\$5.10 for 15 hrs Spark compute runtime on 4-core node.
Analytics Reporting	\$28.00 (Power BI Pro)	\$28.00 (Power BI Pro)	Identical: 1 authoring + 1 consumption. +\$14/mo per new seat.
Total Monthly Cost	\$32.57 / mo	\$39.26 / mo	+\$6.69 / mo net increase for full Lakehouse capability

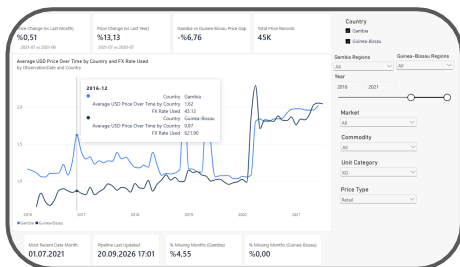
[Link to full pricing document](#)

Service Level Agreement (SLA) Overview

[Link to Document](#)

Priority	Definition	Examples	First Response	Issue Updates	Escalation
1	Dashboard unreachable to all viewers, or the monthly pipeline fails to complete and no refreshed data reaches the dashboard	· Power BI workspace is not updated in time (as per req: ADP-04) · Azure SQL Database does not reflect latest monthly data (as per req: ADP-01) · file, visual exporting not available (.xlsx, csv) (as per REP-03)	Within 4 business hours of detection or notification	Every business day until resolved	Team lead notified within 4 business hours. Fix target: 1 business day
2	A core functionality is broken; rest of the dashboard still works	· One country's data missing or outdated (as per req: ADP-02) · commodity outdated (as per req: FEW-01) · cross-country comparison broken (as per req MON-05)	Within 1 business day	Every 2 business days, or on an agreed interval	Automatic review if unresolved within 3 business days. Assigned to the team member who owns that pipeline stage.
3	Cosmetic or non-blocking issue; a workaround exists	· Currency Standardisation outdated (as per req: as per RCD-05) · minor filter/slicer bug (as per req: REP-01) · slow PowerBI performance (as per req: REP-04)	Within 2 business days	Weekly, or bundled into the next scheduled release	Logged for the next scheduled release. Fix target: next release
Scope Change	Request falls outside the agreed design is logged	· Add a new country or commodity · schema change · new chart type · extend forecasting to a new series	Acknowledged within 1 working day	Periodic update once scoped and scheduled	Routed to the Change Management process (CHG-01): scoped, timed, written sign-off required before building

Conclusion



Working Prototype
(\$0.09)



Affordable MVP
(\$4.57)



Scalable Solution
(\$11.26)